

Original article

Technical and economic feasibility assessment of hybrid renewable energy system at Kenyan institutional building: A case study

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ABSTRACT

This study focused on the technical and economic feasibility assessment of hybrid renewable energy system for powering institutional building in Kenya – a case study of the School of Engineering, University of Nairobi. The study aimed to determine the optimum hybrid renewable energy system that can supply reliable and affordable power to the school. The load profile of the site was determined using power and energy logger (PEL) with data analysis carried out in DataView™ software. The total energy demand and consumption of the establishment were found to be 84.59 kW and 1172 kWh/day, respectively. Grid/wind/photovoltaic/diesel/battery/converter configuration was simulated and optimized in HOMER™ Pro microgrid analysis tool using local meteorological data. The total net present cost was calculated for 25 years to find the lowest cost of energy for the systems. Grid/Photovoltaic/Diesel was found as the optimum system configuration with the system having Cost of Energy (COE) of Ksh 7.89, Net Present Cost (NPC) of Ksh 69,512,100.00, Initial capital cost of Ksh 30,264,100.00, a renewable fraction of 71.6%, simple payback period of six years, discounted payback period of eight years, present worth of Ksh 12,819,300.00, annual worth of Ksh 965,700.00, excess electricity generation of 2.48% etcetera. This result shows that the system is technically and economically viable. The system proved to have reduced the power purchase from the grid by approximately 77% and reduce the electricity bill by approximately 84%. In addition, nine (9) sensitivity cases were used for sensitivity analysis and the result gave the same configuration except in cases where the sizes and prices of some components were changed.

List of Acronyms

UNSDGs	United Nations Sustainable Development Goals
SDGs	Sustainable Development Goals
OECD	Organization for Economic Cooperation and Development
ERC	Energy Regulatory Commission
UNDESA	United Nations Department of Economic and Social Affairs
IEA	International Energy Agency
CF	Capacity Factor
LF	Load Following
CC	Cycle Charging
PW	Present Worth
AW	Annual Worth
FiT	Feed in Tariff
CIA	Central Intelligence Agency
HOMER	Hybrid Optimization of Multiple Energy Resources
NASA	National Aeronautics and Space Administration
NGO	Non-Governmental Organisation
LED	Light Emitting Diode

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List of Acronyms

GIS	Geographical Information System
PV	Photovoltaic
kWh/kW	Kilowatt hour/Kilowatt
TWh	Terawatt hour
GWh	Gigawatt hour
MW	Megawatt
DNI	Direct Normal Irradiation
GHI	Ground Horizontal Irradiation
GDP	Growth Domestic Product
PEL	Power and Energy Logger
PA	Paris Agreement
PPP	Purchasing Power Parity
PPA	Power Purchasing Agreement
PV	Photovoltaic
CO2	Carbon dioxide
KV/KVA	Kilovolts/Kilovolts Amperes

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List of Acronyms	
DC/AC	Direct Current/Alternating Current
DOD	Depth of Discharge
LCOE	Levelized Cost of Energy
COE	Cost of Energy
MPPT	Maximum Power Point Tracker
LPSP	Loss of Power Supply Probability
NPC	Net Present Cost
NPV	Net Present Value
$\beta/\alpha/\lambda$	beta/alpha/lambda

Introduction

The relation between the activities surrounding conventional power generation from fossilized fuels and climate change has become obvious in recent years and scientists and researchers are working on several projects to address the issue. According to the International Energy Agency (IEA), the fluctuation in energy prices, the exponential increase in the world population, and the economic development in third-world countries are the primary reasons for the continuous increase in energy demand [1]. The fluctuation in energy prices, especially in the sub-Saharan Africa, is due to the crippled energy access. Access to energy (heat and electricity) in this region decreased from 980 million to 860 million between 2017 and 2018, with about 600 million people without access to basic electricity [1]. The current global population stands at 7.7 billion and has been projected to increase by 2 billion in the next 30 years, totaling 9.7 billion by 2050 and attain the peak by the end of the century with total of 11 billion [2]. This exponential increase is mainly from Asia and sub-Saharan Africa, with the population of sub-Saharan Africa forecast to double by the end of 2050 [2]. Energy demand globally will continue to increase by 1.3% until 2040 if there are no change in the energy policies and improvement in energy efficiency according to the current policies scenario [1]. Another factor is the concern over the fast depletion of fossil fuels and the urgent need to mitigate the effects of climate change. The stated policies scenario of the IEA indicated that in 2018 the carbon dioxide emission related to energy increased by 1.9% and is projected to reach approximately 36 gigatons by the end of 2050 (World Energy Outlook, 2019). A major challenge for all scientists and researchers remains to meet the increasing energy demand without aggravating climate change, and this is in line with the Paris Agreement (PA) to limit the global average temperature below 2 °C, preferably 1.5 °C. To accomplish the target of the PA, the emissions relating to energy production and consumption needs to decrease to zero by 2050 [1]. The remedy to electrical energy conundrum has been over the years identified as the implementation of more energy efficient systems and the installation of renewable energy as alternative energy solutions [3]. Renewable energy is the energy that is replenished by a natural process at the same rate or faster than the rate which they are consumed. These energy sources include solar, wind, geothermal, biomass, ocean tides, and many more. Solar energy is basically the energy from the sun and many technologies have been developed to harness this “unlimited” energy source including the solar photovoltaic, solar thermal, and wind turbines. These energy technologies improve global economy by providing electricity access to residential and commercial buildings where electricity from the national grid is unavailable or where it is available but is either expensive or unreliable [4].

The present study generally focused on assessing the technical and economic feasibility of hybrid renewable energy system at institutional building in Kenya. Kenya's access rate to electricity stands at 73%, indicating that 27% of the population are yet to be connected to electricity [5]. The current total installed capacity of electrical energy in Kenya is 2336.4 MW, with an effective installed capacity of 2270 MW [6]. The peak power demand rose from 1586 MW in 2016 to 1656 MW in 2017 [5]. This demand is reported to have increased to 1802 MW in

June 2018 [7]. Electricity generation in Kenya is mostly obtained from hydro and fossil fuel power plants, with about 36% and 35% of the installed capacity, respectively, followed by geothermal with about 27%, with no installed capacity from solar energy in the country's energy mix [6], despite the country's potential for solar energy – abundant space for installations and average daily solar irradiance between 4 and 6 kWh/m² [8]. This rate of solar insolation is high with the mean peak sunshine hours ranging from 5 to 7 [9], and about 10 – 14% of the energy from sunlight in this region can be converted to electricity depending on the PV cell efficiency [10]. Although currently Kenya does not have solar energy in its energy mix, small amount of this energy has been tapped in the form of solar stand-alone systems [10] with about 220 schools supplied with solar stand-alone systems accounting for a total capacity of about 574 kW [9]. At the time of this study, tender has been passed for a more institutions of about 117 in number to be supplied power through standalone solar systems. Moreover, 10 million Euros were also provided to the country by Spain to supply solar power to selected 380 institutions in 2017 [6]. These ongoing projects are because of the high solar potential in Kenya. The country is also blessed with fierce wind for power generation. The potential output is 22,476 TWh/year, depending on the turbine capacity factor [10]. Due to its landscape, some areas in Kenya possess outstanding wind speed. The wind speed in Kenya ranges from 5 to 9 m/s depending on the location [11]. The wind speed is greatest at the northwest and some parts in the rift valley with mean wind speed standing at above 9 m/s at a height of 50 m. At this height, the wind speed is lowest in the coast, ranging between 5 and 7 m/s [11,12].

The School of Engineering, University of Nairobi is in Kenya on the Latitude of 1°16.7'S and Longitude 36°49.1'E. The location as obtained from the Google earth map is shown in Fig. 1.3. The school has provided quality engineering teaching and research since its establishment in 1956 under the Royal Technical College, consists of five departments - Electrical and Information Engineering, Civil and Construction Engineering, Mechanical and Manufacturing Engineering, Environmental and Bio-systems Engineering, Geospatial and Space Technology and the Institute of Nuclear Science and Technology. Power is supplied to school from KPLC transformer which has the mains located at the School powerhouse. The power from this mains supply power to all laboratories and workshops together with other sections of the school, including the office of the dean, transport section, estate management block, maintenance and construction section, school Sacco section. Fig. 1 shows the Google earth view of the school. Most of the power is consumed by the equipment in laboratories, workshops, offices and through lighting.

The cost of electricity at this site is considered very high resulting in the school seeking alternative energy sources as a solution to alleviate the present situation. The maximum power demand is given as 121 kW with an average monthly and annual electricity bill of Ksh. 552,155.46 and Ksh. 6,625,865.52 respectively as observed from the energy bill, although the study carried out one-month monitoring and measurement for validation. The objective of this study was to determine and propose an optimal grid-tied hybrid renewable energy system which can supply stable, inexpensive, and dependable electricity to the school of Engineering, University of Nairobi. To comparatively assess the feasibilities of hybrid renewable energy systems (PV/Wind/Diesel generator) for reliable energy supply to this establishment. This goal was realized by achieving objectives which include: determining the daily load profile of the site and the examination of the renewable energy potential in this area; identifying the appropriate renewable energy resources for the suggested hybrid system and its constituents selection and cost analysis; modelling and simulation of the system using a computer software package and finally optimization and sensitivity analysis of the hybrid system and performance assessment of the best hybrid system.

Literature review

Many studies have been done to compensate for a standalone, grid-



Fig. 1. School of engineering (Google Earth).

tioned and hybrid renewable energy systems to provide alternative energy solutions for high electricity reliability and affordability around the globe. The aim of this review is to justify the theoretical framework and the methodology utilized in this study. The comparison of electricity cost generated from a fuel-cell based RE and the electricity from the national grid was carried out in [13] for powering a health center in rural India with the aim to evaluate the cost of decentralized RE system and compare it with the national grid. The result of the study showed that the decentralized RE system is feasible and sustainable for this location. Apart from the electricity generation through fuel-cell-based electricity solution, in India, an off-grid electricity power generation was carried out using the application of HOMERTM software in [14]. The study aimed to propose the best technology configuration for combined renewable energy sources, including small hydropower, PV systems, wind turbine and a biodiesel backup generator. The result was positive by indicating that the off-grid renewable energy system is feasible within the examined location. In Malaysia, analysis of hybrid photovoltaic-diesel energy system was carried out in [15]. The study examined the potential of electricity generation in different location of the country under the country's climatic conditions and analyzed the economic feasibility with the use of HOMERTM energy simulation tools and based on different solar irradiances and diesel prices. It was established that hybrid renewable energy systems are cost-effective within the regions. Techno-economic analysis of hybrid renewable energy system were also carried out in Cambodia and Egypt [16;17] while hybrid system comprising grid component was studied in Saudi Arabia in [18]. HOMERTM tool was also utilized in the above studies to determine the feasibilities of the systems which proved to be positive. Studies in this area also extended to integration of biogas into PV-wind hybrid renewable energy system which had proven that mainstreaming biogas in hybrid energy systems can help in the adoption of alternative energy

solution. A case study carried out in [19] examined the technical and economic feasibilities of this system in a commercial building in India using mathematical models and optimization. The result indicated that both the capacity factor and the percentage of energy generation was highest on the biogas side and the system showed technically and economically feasible. Solar-biomass off-grid energy system was also carried out in a rural area of Pakistan in [20,21] to evaluate the economic and technical feasibility of the system in HOMERTM energy tool for provision of electricity in this region. The Net Present Cost (NPC) and Cost of Electricity (COE) of the study showed that the system is feasible for this location. Hybrid solar thermal technology has also been studied as shown in [22]. This study, through experimental design and mathematical models, established the economic feasibility of a PVT system consisting of a solar dish concentrator. The result showed that the levelized cost of heat (LCOH) was competitive when compared with conventional electric heating, natural gas, and LPG. Two decentralized power systems with different configurations in Malaysia were analyzed in [23] to investigate their performance characteristics considering the combination of PV/Diesel/Battery technologies using HOMER for modeling and optimization. The authors based their investigation on the technical, financial, and environmental issues while modeling all possible scenarios of various configurations using data specifically from the site. The study showed that the hybrid system configuration of PV/Diesel/Battery, for the same load profile and weather data provides 24 h electricity access and gives the best performance when compared with other scenarios in technical terms while the scenario of diesel generator alone gives the best energy supply economically. The study further showed that 100% renewable energy is not economically feasible due to the high capital cost of battery storage system and its replacement cost although this is likely to change due to progressive research being carried out in this area. Investigation of the performance of an off-grid

hybrid system of PV/Wind/Diesel/Battery located at the South China sea, Malaysia was studied using HOMER tool for the technical and economic evaluation in [24]. The weather data was specifically for the site and was acquired from the NASA surface meteorological and solar database. The result shows that Hybrid Renewable Energy System (HRES) is economically feasible based on the NPC economic criteria and enhances significant reduction in the CO₂ and other greenhouse gas emissions into the environment. Recognizing the challenges in grid expansion to some rural areas in Nigeria, [25] investigated the economic assessment of HES for electrification of rural areas in the six geopolitical zones of the country. The study aimed to evaluate the techno-economic viability of hybrid energy systems comprising of PV/Wind/Diesel using HOMER energy software in these rural locations and as such, obtain the optimum solutions for each location. The authors established the load profile by assuming typical domestic energy consumption and social infrastructure for two seasons existing in Nigeria and made use of weather data from NASA while simulating three sensitivity cases for the diesel price. Based on the economic criteria, NPC, COE and Renewable Fraction, the study found that the PV/Diesel/Battery is the optimum configuration in all the six geopolitical zones examined while the diesel standalone system showed to emit significant amount of CO₂ into the environment and proved to be more expensive in terms of the COE. Further, the study showed that wind/diesel system is not economically viable due to insufficient wind speed in the sites. Analysis of long-term weather data for hybrid renewable electricity application in rural health clinics (RHC) was also investigated to ascertain the technical and economic feasibility of the systems in RHC spots by [26]. The study was carried out in different locations of the six geopolitical zones in Nigeria using HOMER energy for analysis of daily weather data ranging between 18 and 39 years for PV/Wind/Diesel/Battery hybrid system. The result showed that monthly average and the annual average power density of the sites as well as the optimal tilt angle vary for the locations although in general the study showed that integration of wind and solar energy into standalone systems in these locations are feasible for improving electricity supply in RHC in Nigeria. The PV/Wind/Diesel/Battery hybrid system was found to be feasible and optimum for RHC in some locations while PV/Diesel/Battery showed to be the best configuration for RHC in some other locations. This inconsistency was found to be as a result of the variation in weather data for these six locations.

A review of the literature has shown that a combination of different renewable energy sources for power generation as well as diesel generator and battery storage system can provide solution to grid instability and supply power to rural areas and suburbs. However, different locations have varying renewable energy potential and meteorological data therefore the feasibility of HRES is not guaranteed technically and economically. The results obtained in one location cannot be used to generalize the feasibility of the system; hence the need for investigation of different sites to ascertain the viability of the system. The methodology for carrying out such investigation is also important as different approaches are bound to give dissimilar results. Many studies focused on the standalone (off-grid) system capitalizing on the expensive nature of extending national grid to the rural areas and the difficulty in handling the terrains of the locations. However, the present study is a case study of institutional building in Nairobi, Kenya and is aimed to investigate the technical and economic feasibility of the HRES in this institution. In this location, grid extension is in existence therefore the investigation was to find the optimal solution to the grid instability by integrating renewable energy sources into the existing grid. In this regard, Grid/Wind/PV/Diesel/Battery system has been modeled, incorporating other system components such as the two-way inverter for optimum solution. Compared with previous studies [25,26], the study took into consideration the existing grid extension, FiT policy and the power purchasing agreement (PPA) of the site location to focused on finding the best system solution both technically and economically while these studies focused on the viability of standalone systems. The present study also differs since it uses quantitative research approach in

collecting the data for establishment of the load profile of the site.

Limitations and assumptions

The proposed approach for finding the optimal HRES for powering the institutional building in Nairobi, Kenya has various limitations as well as different assumptions that were made to accomplish the objectives. First, the study is specifically for the selected location (Latitude of 1°16.7'S and Longitude 36°49.1'E) and is not the best configuration for other locations where renewable energy potential is not the same as the investigated site even if they have the same load profile. Renewable energy such as biomass was not considered due to its unavailability at the site hence the system is specifically for solar and wind energy which are available with other components (generator and grid) with the storage batteries and two-way converter. In addition, the study does not address the complete design of the micro-grid powered by this HRES and the voltage, current and frequency stability because of the integration of renewable energy sources into the grid were not considered but is recommended for future study. Further to the limitations, several assumptions were made. The inflation rate in Kenya was assumed to be the same throughout the lifetime of the project and the data from NASA meteorological database were assumed to be sufficiently accurate for estimating the solar and wind energy resources as well as the temperature of the site. The study also assumed that the annual dissimilarities for solar irradiation and wind speeds occurring throughout the project lifetime remains the same.

Methodology

The research method used to establish the energy demand and consumption data of the establishment primarily consists of the following three approaches:

Obtaining the energy demand and consumption of the establishment from Kenya Power and Lighting Company (KPLC) for at least one-year period in order to identify the suitable period (peak month) for monitoring; studying the energy equipment used by the establishment to obtain their performance characteristics and the use of Power and Energy Logger (PEL) to monitor the electrical power demand and consumption of the establishment from the mains during the peak month.

These approaches were used to establish the load profile of the school. Energy demand and consumption were monitored for a one-month period (peak month), using the PEL 102/103 power and energy online logger. The first two weeks were during the exam period when several activities such as printing, photocopying, night studies by students, etc. were on in the school. The second two weeks were after the exam, during the fourth session of the school when activities were carried out daily in the laboratories and workshops. The data obtained using PEL was analyzed using the DataView™ software. In addition, the renewable energy potential of the site was determined using HOMER software by inputting the geographical coordinates of the site since ground measured data were unavailable. To evaluate the effect of uncertainties, nine (9) sensitivity cases were carried out with a focus on the change in the price of diesel fuel and the weather data. The system components were selected alongside their costs from both local and international vendors for both technical and economic optimizations. The renewable energy data of the site, load profile, system components (PV, wind turbine, grid, batteries, generators, charge controllers, inverters, etc.) and their respective costs were also modeled on the software interface, economic constraints such as the inflation rate and discount rate and the geographical coordinates were inputs to the HOMER™ Pro Micro-grid Analysis Tool (version x64.3.12.3) for simulation and optimization. The time taken to complete simulation was dependent on the values of sensitivity variables specified and the result of the simulation ranked in the software based on the economic criteria. Fig. 2 illustrates the summary of the fundamental input-output data in HOMER™ software. Two optimization algorithms are used in HOMER –

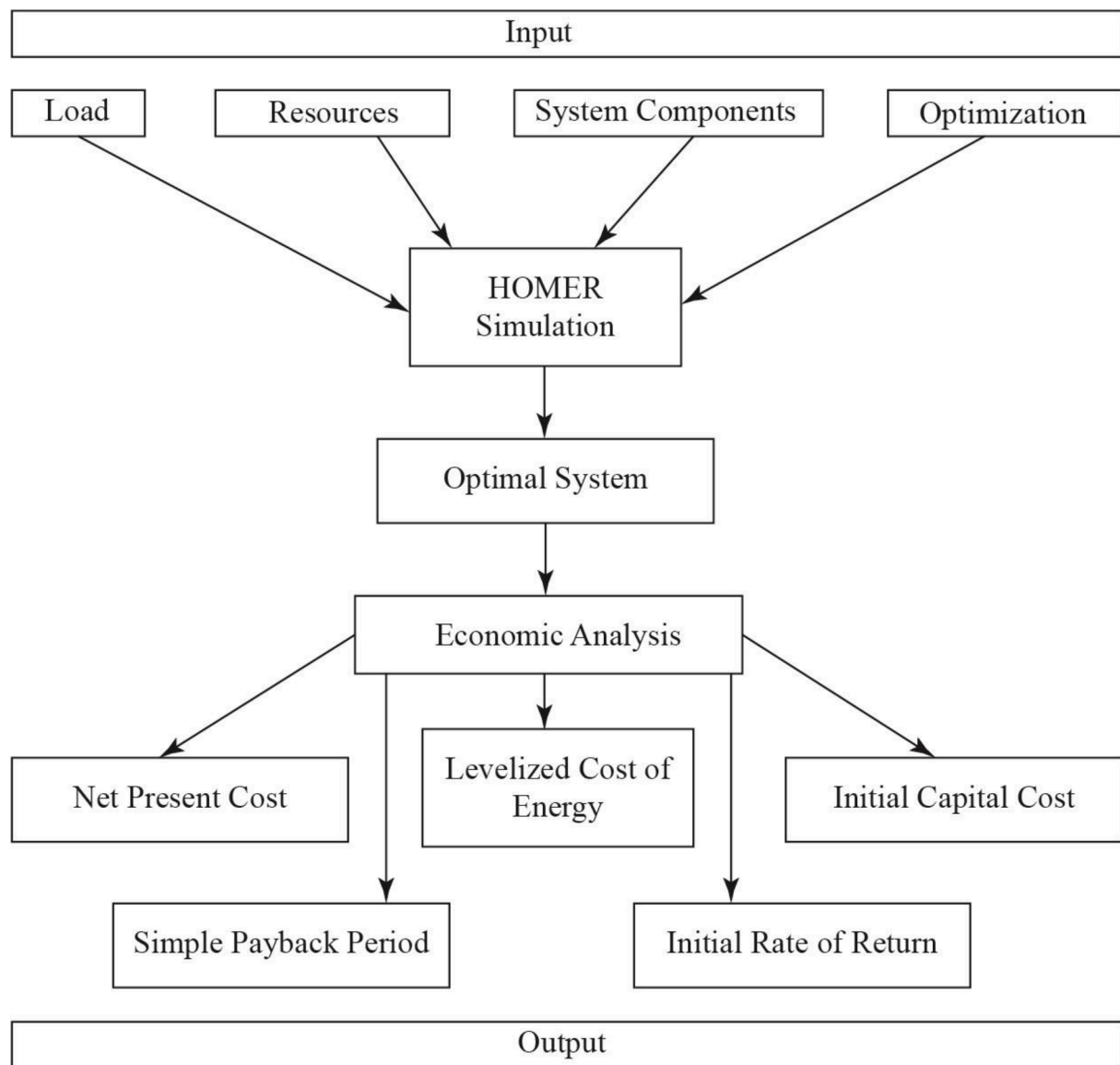


Fig. 2. HOMER input–output parameters.

the original grid search algorithm (OGSA) and the proprietary derivative-free algorithm (PDA). The HOMER OGSA considers all the system configurations specified in the search space that are feasible and simulate them for optimal results while it uses PDA to search for the most economical system which are sorted in the optimization result

table based on the net present cost for design option comparison. HOMER uses PDA to rank the system configurations based on the economic criteria and uses OGSA to examine the feasibilities of the system configuration as specified in search space and the sensitivity variables. HOMER repeat the optimization process for every sensitivity variable

Table 1
System components, parameters, and their values.

Component	Parameter	Value	Component	Parameter	Value
Wind Turbine	Rated Power	10 kW	PV	Nominal Max Power	355 W
	Maximum Output Power	13 kW		Optimum Operating Voltage	39.4 V
	Start-up Wind Speed	2.5 m/s		Optimum Operating Current	9.02A
	Rated wind speed	10 m/s		Open Circuit Voltage	46.8 V
	Survival Wind Speed	25 m/s		Short Circuit Current	9.59A
	Swept Area	50.2 m ²		Module Efficiency	17.89%
	Rotor Speed	8 m	Battery	Capacity	840Ah
Inverter/Converter	Rotor Blade Diameter	180 rpm		Nominal Voltage	6.91kWh
	Nominal DC input voltage	48VDC		Nominal Capacity	1156Ah
	DC Input Voltage Gauge	40 – 64VDC		Maximum Capacity	40%
	Max. AC input current	50AAC @240 ADC		Minimum State of Charge	41A
	Typical Efficiency	93%		Round Trip Efficiency	80%
	Capacity	8 kW			
	Grid Interaction Voltage Range	106 – 132VAC			

defined as an input.

System components and specifications

Solar photovoltaic (PV)

The physical examination of the site shows that the PV panel installation can be done on the roof tops of the buildings on a fixed axis, at an angle that is equivalent to the site latitude for optimum solar radiation capturing. In this research, 355W CS3U-355P-40MM poly-crystalline Canadian solar had been selected. The essential specification of this PV is shown in Table 1. The cost of this panel is \$0.50 per Watt and \$180.00 per panel from [27] as at 7th January 2019. The cost of shipping to Kenya is estimated by the seller to be \$5.00 per panel while the cost of local transportation to the site and installation is estimated to be \$20.00 per panel. The total capital cost is therefore \$225.00 per panel and a replacement cost of \$0.00 per panel since the lifetime of the project life is equal to the lifespan of the PV panels. To take care of the declination in efficiency of the PV cells due to the fluctuation in temperature, accumulation of dusts and losses due to wiring, the study included a derating factor of 0.8. The PV derating factor has been established ranging between 0.7 and 0.8. Ground reflectance is also accounted for in this study. Ground reflectance is the part of solar radiation incident on the ground which reflects to the atmosphere that is dependent on nature of the ground. The value of ground reflectance ranges between 20 and 70% [28]. Because at the site, there are no such things as snow and the dust accumulation is negligible, 20% had been used as the ground reflectance in this present study.

Storage battery

Surrette 6CS-25PS battery was selected for this study. The total capital cost including the shipment and installation cost is per one. Its properties are given in Table 1 and the quantity required to meet the maximum load specified as 0, 150, 200, and 300 in HOMER search space for selection of the appropriate number for simulation and optimization. The battery cost \$1,200 per piece with the total capital cost estimated as \$1,200.

Wind turbine

The wind turbine is expected to generate enough electricity to make a significant contribution to the renewable fraction although this depends largely on the wind speed potential. In selection of a wind turbine, factors such as the number of turbines, height of hub, service times, costs type of electricity produced (AC or DC) and the cut-in wind speed are all considered. In this study, 10 kW Aeolos wind turbine had been selected putting the wind speed potential of the site into consideration. The reference height (hub height) of 20 m was used for optimization although Aeolos 12 m and 18 m hydraulic lift tower is available for the 10 kW wind turbine as the time of the study. Essential parameters of the wind turbine and their values are given in Table 1. The cost of Aeolos 10 kW wind turbine as at January 2019 is approximately \$20,000 per assemble. Cost of shipping from United Kingdom to Kenya is estimated to be \$1,200 with local transportation estimated to be \$300. Installation cost is estimated to be \$5,000 while O & M cost is taken to be \$500/year. The total capital cost of this wind turbine is therefore rounded up to \$27,000 to accommodate miscellaneous costs. The replacement cost is taken as \$22,000. For quantity (number) consideration of this wind turbine, quantities 0, 2, 4, and 6 had been specified for reliable optimization. Depending on how fierce the wind speed is, HOMER will select the appropriate quantity to give a maximum result.

Converter

The inverter should be rated the same capacity as the peak load or higher but because in this design, the power is supplied by both renewable and non-renewable power sources, the inverter with the same capacity as the peak demand is selected. O&M costs for this equipment were neglected since they have little or no maintenance when installed.

In this study, OutBack 8kw 48 V pure waveform hybrid inverter has been selected [29]. The important specifications of this inverter are given in Table 1. Converter sizes considered are: 0, 50, 100, 200 and 250 kW for optimization; Capital cost of converter is taken as \$4,500 per 8 kW and the replacement cost is about \$4,000. The typical efficiency of the converter is at 93% with a lifetime of 15 years.

Dispatch strategy

There are some standards and through which the generator and the batteries must adhere to whenever they must supply power. This standard is referred to as dispatch strategy. In this study, the only dispatchable energy sources used are the diesel generators. Power from the renewable energy sources is used to supply the power demand but when in excess, they are used to charge the batteries. When the power generated from renewable energy power sources is low, the generator does the work of charging the batteries. At times it is more economic viable to operate the generator at its full capacity when needed and use excess power to charge the batteries. In HOMER energy software, two dispatch strategies are used:

Cycle Charging (CC): When power from generator is needed, it is operated at its full capacity with the excess power being used to charge the batteries. This strategy is suitable for a system with small or no renewable energy power sources. It is represented with the acronym CC in the result sheet.

Load Following (LF): When power from the generator is needed, it is operated in less than its full capacity to meet the load demand. The strategy is suitable where power from renewable energy sources is considerably high. It is denoted with the acronym LF in the result sheet. In this study, Load Following was adopted.

Grid

For proper and accurate investigation, both grid-tied and off-grid hybrid system are accounted for in this study. The Grid page in HOMER allows specification of grid in many ways which includes:

Simple rates mode allows specification of site's constant but average electricity cost per kWh, the sell back cost, and the capacity that can be sold back. This rate was adopted in this study. The analysis of the 13 months electricity bills of the site showed that the peak energy tariff is approximately Kshs 20.00 (\$0.20) and this cost specified in HOMER for optimization. Connecting the system to grid means that at some points, excess power would be generated by the renewable energy sources and when this happens, the extra power is sold back to the grid. Since Feed in Tariff (FiT) policy in Kenya enables power producer to sell back to the grid at a maximum price of Ksh 12.00 (\$0.12) per kWh, this value was specified during the simulation.

Real time rates mode which describes costs on an hour basis through introducing accurately configured text file which has time-series data. This is usually suitable for advanced grid modelling and therefore was not adopted for this study

Scheduled rates mode which permits different costs of power at every time of day as well as every month of the year to be specified.

Grid extension mode which allows the comparison of the price of grid extension and the price of every standalone power system in the system arrangement.

The concept planned behind this project is that the institution would sign the PPA with the KPLC, which would permit the power trade with the national grid at the maximum rate of \$0.12/kWh as given by the policy and have been done [30]:

Economic criteria

The total Net Present Cost (NPC): This is the difference between the present value of the total costs which the system incurs throughout its lifetime and the present value of the revenue from the system throughout its lifetime. The present value of the costs that the system will incur in n -years later can be estimated using Eq. (3.1)

$$C_{NPC} = C \left[\frac{1+i'}{1+d} \right]^n \quad (3.1)$$

where d is the nominal interest rate and i' is the annual inflation rate, both in percent (%).

To determine the Levelized Cost of Energy (LCOE), the overall NPC of the system has to be definitely converted to sequence of same yearly cash flows which is referred to as overall annualized cost. Eq. (4.16) can be used to deduce the overall annualized cost of the system:

$$\text{Total annualized cost (\$/year)} = \text{Total NPC} \times \text{CPF} \quad (3.2)$$

Where, CPF equal to capital recovery factor which is defines the ratio utilized in estimating the present value of sequence of yearly cash flows. The CPF can be calculated using Eq. (3.3):

$$\text{Capital Recovery Factor} = \frac{i(1+i)^N}{(1+i)^N - 1} \quad (3.3)$$

where N = the number of years (equivalent to the lifetime of the entire project) and i = real interest rate.

HOMER computes the yearly real discount rate the “Nominal discount rate” and “Expected inflation rate” inputs. It makes use of real interest rate to compute discount factors together with the annualized costs from the project’s net present costs. Eq. (3.4) gives the real discount rate:

$$i = \frac{d - i'}{1 + i'} \quad (3.4)$$

Discount factor, f_d is estimated in HOMER using the expression given in Eq. (3.5):

$$f_d = \frac{1}{(1+i)^N} \quad (3.5)$$

Kenya inflation rate fluctuates. As at November 2018, it is 5.58% with annual rate of 4.59% however, it has been projected to 6.5% therefore; the inflation rate of 7% had been used in this simulation. The discount rate is kept constant at 9% by the CBK. It means that Kenya can borrow at 13% [31]. In this regard, the discount rate of 13% had been chosen in this simulation.

Annualized system cost (ASC)

ACS is the addition of the annualized cost (capital, replacement, and maintenance) of all the system components.

$$\text{ASC} = \text{Annualised}(\text{capital} + \text{replacement} + \text{maintenance})\text{costs} \quad (3.6)$$

Levelized cost of energy (LCOE)

LCOE evaluates the system considering the overall recurring and non-recurring costs throughout the lifespan of the project. It is the ratio of the overall ASC to the annual electrical energy production (E_{Total}) in kWh. Mathematically,

$$\text{LCOE} = \frac{\text{TotalAnnualisedcost(USD/yr)}}{\text{AnnualLoadServed(kWh/yr)}} = \frac{\text{ASC}}{E_{\text{Total}}} \quad (3.7)$$

Internal rate of return

The IRR is sometimes referred to as the return on investment (ROI) or time adjusted rate of return. It is the interests earned by the system throughout its lifespan. This is evaluated by setting the project NPV value to zero and then compute the discount rate.

Payback period (PBP)

This is the period during which the original investment made on the system is recovered from the cash inflow generated from the project. It is given as:

$$\text{PBP} = \frac{\text{InitialInvestment}}{\text{CashFlowperperiod}} \quad (3.8)$$

Search space

To find the optimum system architecture, search space in HOMER is utilized. This is a space in HOMER where component capacities and/or quantities are specified to enable the software select and optimize the best capacity and/or quantity. Table 2 shows the search space for this system.

Sensitivity variable

Sensitivity variables are variables specified in HOMER to account for what happens when different parameters such as wind speed, solar irradiation, temperature, diesel price etc. changes in the future. HOMER carries out simulations and optimization separately for each of the sensitivity variable specified. Table 3 gives the sensitivity cases performed in this study

Result and discussion

Figs. 3 and 4 show the energy demand pattern of the case study building during the first two weeks and the second two weeks of energy monitoring. The peak demand has been identified to be 84.59 kW and 82.56 kW during the first two weeks and second two weeks, respectively.

Table 4 shows the daily peak demand and consumption while Fig. 8 shows the hourly power demand for the baseline period. The day with the highest peak demand in kW was selected to obtain the hourly load profile of the establishment. Fig. 5 shows the demand per minute during the peak day over the baseline period. From Table 4, the peak demand and the consumption for the baseline period is 84.59 kW and 1172 kWh, respectively.

Fig. 5 shows the one-minute summary of the daily demand for the peak day. The site exhibits bell-curved demand pattern showing how the demand rises from morning, picking up through the afternoon and possesses its peak at around 12 noon. There was a significant decline in the demand between 1.00 pm and 2.00 pm as a result of the break time allocated for lunch. The demand curve then slows downwards as evening approaches and the workers retire to their homes.

The identified suitable renewable energy resources for the proposed system

As a result of the unavailability of other renewable energy resources in the site, the study focused on the Variable Renewable Energy Sources (VRES) – wind and solar. The potential of these resources was examined in section 1 of this study. The data for plotting the mean speed of wind at the site and the solar irradiation as well as the site temperature were obtained from the NASA meteorological database (<http://eosweb.larc.nasa.gov/>) [32]. Fig. 6 gives the daily mean solar irradiation incidents on a horizontal surface and their corresponding clearness index throughout the year. This value is highest during the month of February with a daily solar irradiation and clearness index of 6.68 kWh/m²/day and 0.655, respectively. This is a period when the weather tends to be hotter with the sun radiation at its peak. The mean of these values was found giving that the annual average solar irradiation is 5.93 kWh/m²/

Table 2
HOMER Search Space.

Converter Capacity (kW)	Generator Capacity (kW)	Grid Purchase Capacity (kW)	PV size (kW)	Surr6CS25P Strings (Number)	10 kW Wind Turbine (quantity)
0	0	(default) 999,999	0	0	0
50	360	–	50	150	2
100	–	–	100	200	4
200	–	–	200	300	6
250	–	–	250	–	–
300	–	–	300	–	–

Table 3
Sensitivity Cases.

Diesel fuel price (USD/L)	Solar scaled average (kW/m ² /day)	Temperature Scaled average (°C)	Wind scaled average (m/s)
0.9	5.93083333	19.8975	4.764166667
1.0	4.931	18.5	3.764
1.2	–	–	–

day, and the standard deviation was found to be 0.55 meaning that the variations of solar irradiation between the months are negligible.

Fig. 7 shows the average monthly wind speed of the site measured 50 m above the surface of the earth. The annual average wind speed was determined to be 4.76 m/s. The wind speed is highest in the month of September with an average value of 5.15 m/s and lowest in December with an average speed of 4.32 m/s. The standard deviation of these values was determined as 0.76 showing that the values are spread apart from the mean and suggests that the power generation by the wind turbine may be minimal.

The monthly average temperature at the site was found between the range of 18.58 °C to 21.04 °C throughout the year as shown in Fig. 8. The temperature experienced during the day and during the night varies between 3 °C and 5 °C. Due to the peculiar topography of the site, the daytime hours all year round does not have a significant deviation from each other and this means that load curve does not have significant deviation for the whole year, which is the reason why the load profile of the peak day observed from data analysis has been used and assumed to be the same throughout the remaining months of the year. Since the energy produced by PV panels is proportional to rooftop area available, the school rooftop areas were measured. The school has abundant roof space for PV installation and ground space for wind turbine installation. The total roof space available for PV installation was found to be 1970 m². The temperature of the site is an important parameter for power generation, although at the present site, it varies between months. In this case study, February has the highest temperature of approximately

21.04 °C while July has the lowest temperature of approximately 18.6 °C as can be observed from Fig. 8. The annual average temperature of the site is calculated in HOMER as 19.90 °C. This temperature is within the design temperature of most fabricated solar PV and would not affect the performance characteristics of the PV power module.

Simulation and optimization

In this section, detail of the simulation and optimization results for the systems under consideration which were required to meet the scaled average load demand of 145 kW peak and 1200 kWh/day consumption were given. The design and optimization were carried out in HOMER™ Pro Micro-grid Analysis Tool (version x64.3.12.3). Based on the economic criteria, the optimum hybrid renewable energy system was selected. This is the system which would be able to meet the load demand at an affordable price and minimized CO₂ emission. The simulation was carried out several times to ensure that a feasible result was obtained, after applying the entire input variables in the modelling tool. The results of the simulation and optimization are shown as total but categorized, with the most feasible system that can supply the peak load demand and the specified constraints ranked on top of the result table. The ranking is done based on the economic criteria and satisfaction of the specified constraints. This entails that the first system on the result table is the best system followed by the next and continues in that order. The economic criteria that were considered for the ranking include, but not limited to NPC, LCOE, Initial Capital Costs, Fuel Costs, O&M costs. Among other criteria include the renewable energy penetration, excess electricity generated and capacity shortage. Fig. 9 shows a schematic of the hybrid system design with all the components (technologies) under consideration.

The following simulation and optimization scenarios in Table 5 are proposed for the detailed analysis that will result in determining the best hybrid system. The system which gives the least Net Present Cost (NPC) and Levelized Cost of Energy (LCOE), least capacity shortage and excess power production, higher renewable energy fraction and lowest fuel consumption at each study site would be chosen as optimal. All the

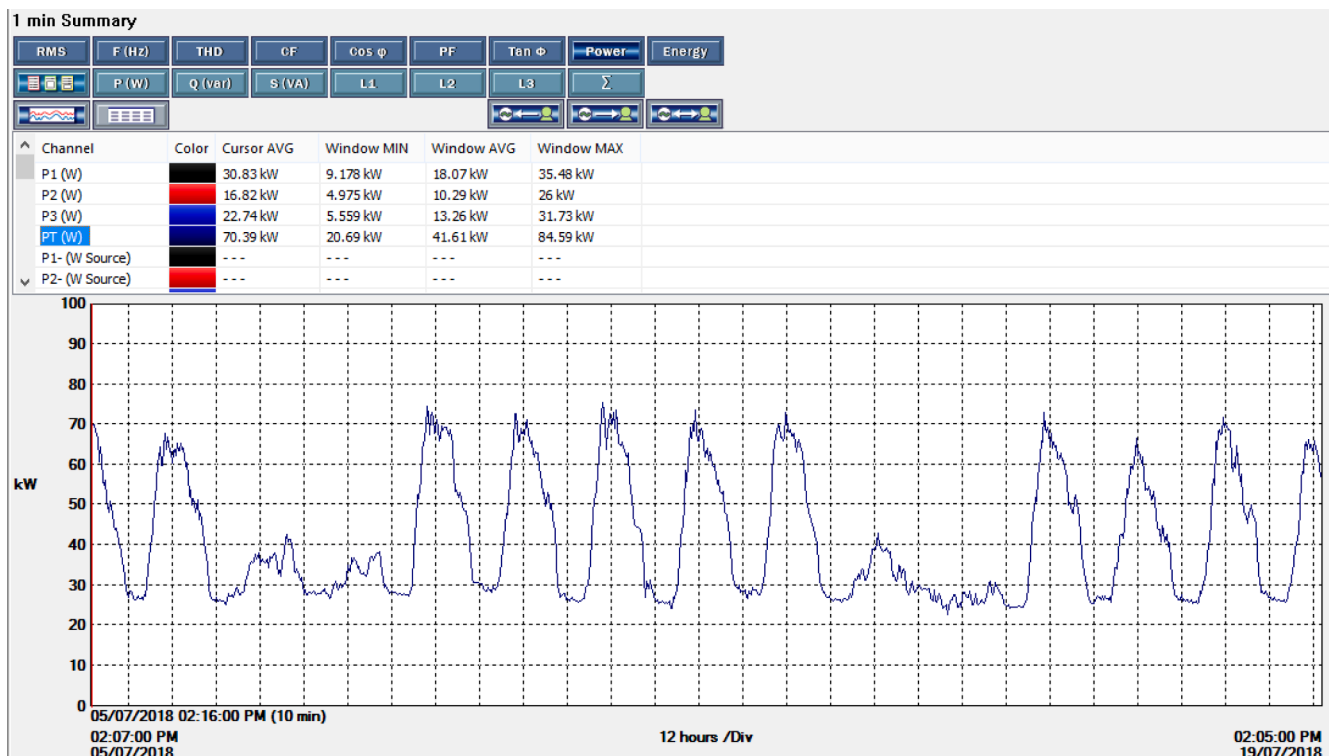


Fig. 3. Energy demand pattern of the school for the first two weeks.

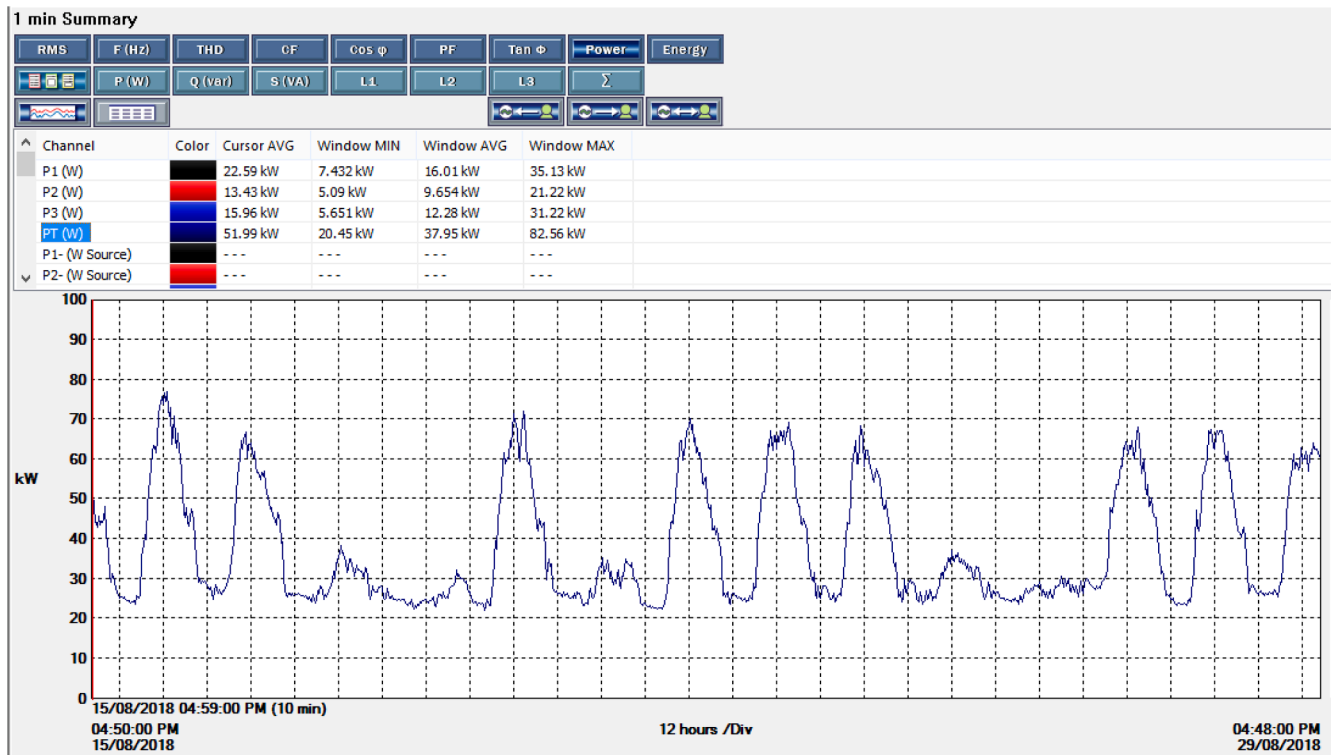


Fig. 4. Energy demand pattern of the school for the second two week.

Table 4

Daily peak demand and consumption of the school for the baseline period.

Date	Peak Demand (kW)	Consumption (kWh)
06/07/2018	82.56	1098.0
07/07/2018	72.43	1049.0
08/07/2018	54.55	755.6
09/07/2018	36.22	626.1
10/07/2018	78.87	970.1
11/07/2018	59.2	748.0
12/07/2018	74.98	971.1
13/07/2018	74.12	1049.0
14/07/2018	72.19	1032.0
15/07/2018	53.86	761.7
16/07/2018	35.98	647.0
17/07/2018	72.4	975.0
18/07/2018	74.89	1051.0
19/07/2018	69.07	1019.0
16/08/2018	74.22	1147.0
17/08/2018	70.98	891.8
18/08/2018	45.45	777.0
19/08/2018	79.09	1020.0
20/08/2018	78.08	1172.0
21/08/2018	81.1	1135.0
22/08/2018	80.59	1087.0
23/08/2018	79.45	1130.0
24/08/2018	70.7	892.0
25/08/2018	41.53	706.3
26/08/2018	76.18	895.4
27/08/2018	72.59	1045.0
28/08/2018	84.59	1035.0
29/08/2018	73.15	1045.0

scenarios were optimized using the same data set. Scenario A consisting of Grid/PV/Diesel generator system (grid-tied) was ranked the best among the systems (A to J) indicating that technically and economically this system is feasible for powering the selected site.

Table 6 shows the selected component capacity and their quantity. As shown in the table the optimization capacity of the PV is selected as 300 kW. This is viable since the rooftop space is enough for this

installation and the excess energy generated can be sold back to the grid or used to supply power to other part of the institution. Two 10 kW wind turbine was selected during optimization, totaling the installed capacity of the wind turbine to 20 kW. Since up to 6 of this 10 kW wind turbines were specified, and HOMER picked two only, it implies that the viability of wind power generation is low due to the low wind speed that exists at the site. 300 pieces of 355 W battery were chosen in other to store surplus power from the solar PV system. Also, 200 kW the two-way converter was utilized to convert DC to AC and vice versa when needed. The charging style chosen was load following which implies that power can be used as the batteries are being charged. Table 7 shows the resource input data as were found. The system was modelled based on the worst-case scenario with varied diesel price at maximum of \$1.2/L. The combined system ranking was done based on the economic viability as mentioned in previous section with system A (Grid/PV/Deisel generator) emerging as the optimal system configuration. Table 5 shows the best systems as they follow each other in ranks.

Optimization analysis of the selected optimum hybrid system

To analyze this system, the components are taken part by part. In general, the pattern of monthly electricity production in kW achieved after simulation is shown in Fig. 10. The solar potential of the site is relatively high, having a maximum in February and a minimum in July. The system produces total electricity of 706,346 kWh/yr, with 663,791 kWh/yr being utilized, leaving an excess of 17,524 kWh/yr (~2.48%). The analysis showed that the system has no unmet load demand and capacity shortage. About 73.4% of the overall energy generated is from the solar PV while the grid contributes only 26.6% of the total energy. Due to the high renewable fraction (71.6%) and maximum renewable penetration of 146%, the diesel generator usage is so minimal to be shown although its presence is required for backup purposes. From Fig. 10, it can be observed that, out of about 85 kW, the grid contributes only about 20 kW. This implies that the system reduces the power being purchased from the grid by approximately 77% (90 kW 20 kW). During the day, the power generated by the solar PV is enough to meet the load

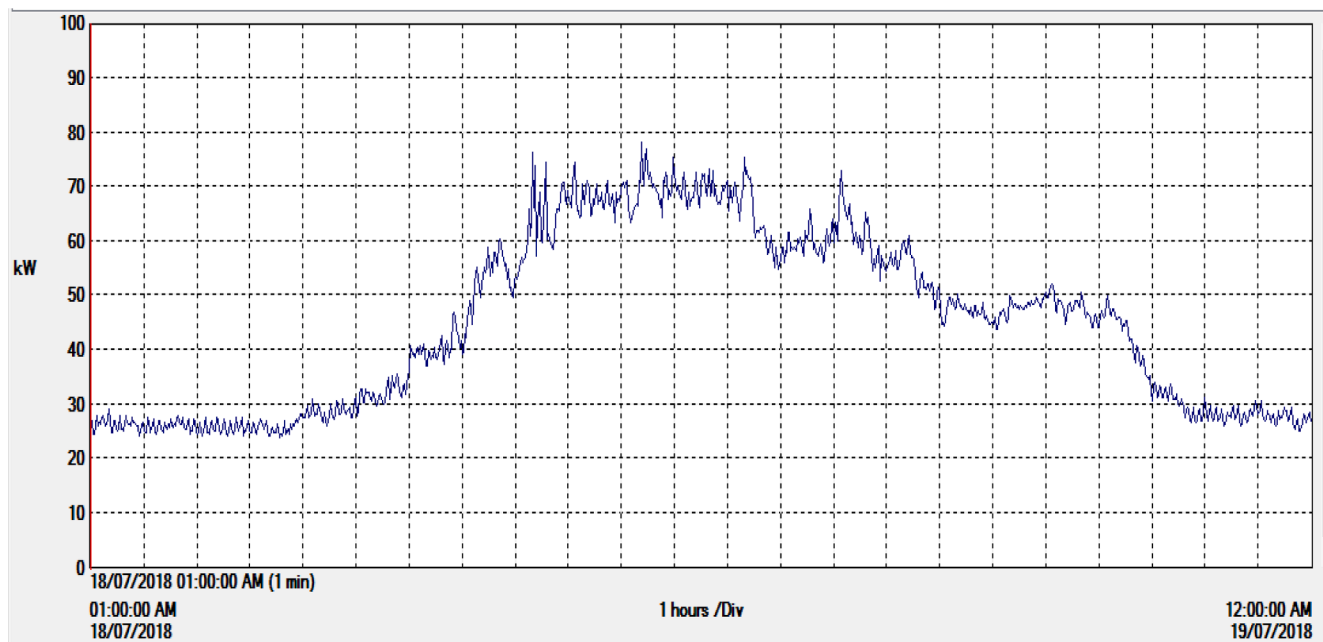


Fig. 5. One-minute summary of energy demand of the site during the peak day.

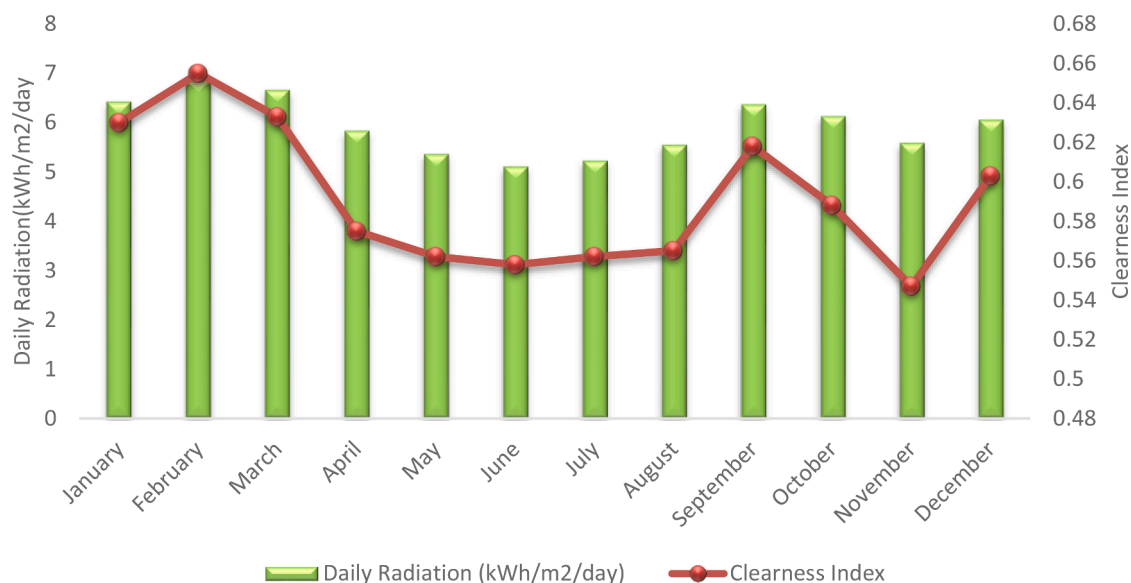


Fig. 6. Daily mean annual solar insolation incidents on a horizontal surface at the site.

demand as can be observed from Fig. 11. The figure shows that between 06:00 and 18:00, much of the electricity usage comes from the solar PV system.

Grid power analysis

For this system, the power from the grid is only used when the power generated from the PV is minimal. Since there is no storage system, the PV only takes care of the power demand during the day while at night, the grid takes over. The power generated by the PV system is sufficient to meet the demand, hence during the day, the net power purchase from the grid is zero. At night, the demand is low, and this is met by purchasing power from the grid. The net energy purchased in most months of the year is negative, giving rise to a zero-demand charge. The energy charge per year is approximately \$10,500 (Ksh. 1,050,000). Decreasing the annual electricity bill of the site by approximately 84%. Compared to

the base system, the system gave an internal rate of return of 15.2% and simple payback and discounted payback period of six (6) years and eight (8) years, respectively. Table 8 shows the monthly power demand bought from the grid and the amount sold back to the grid while Fig. 11 gives more insight about the power market that exists for this system.

Observation of the analysis from Fig. 11 showed that energy is purchased from the grid at night between 7 pm and 7am each day when there is little or no power generated by the renewable energy power generation system. During the day, between 6am and 6 pm, excess energy generated from the renewable energy source is sold to the grid. This process puts the net energy purchased to negative, thereby nullifying the demand charges. The implication of this result entails that the establishment will not be charged for the energy purchased from the grid since the revenue obtained through the sale of energy to the grid is used to pay for the power purchased.

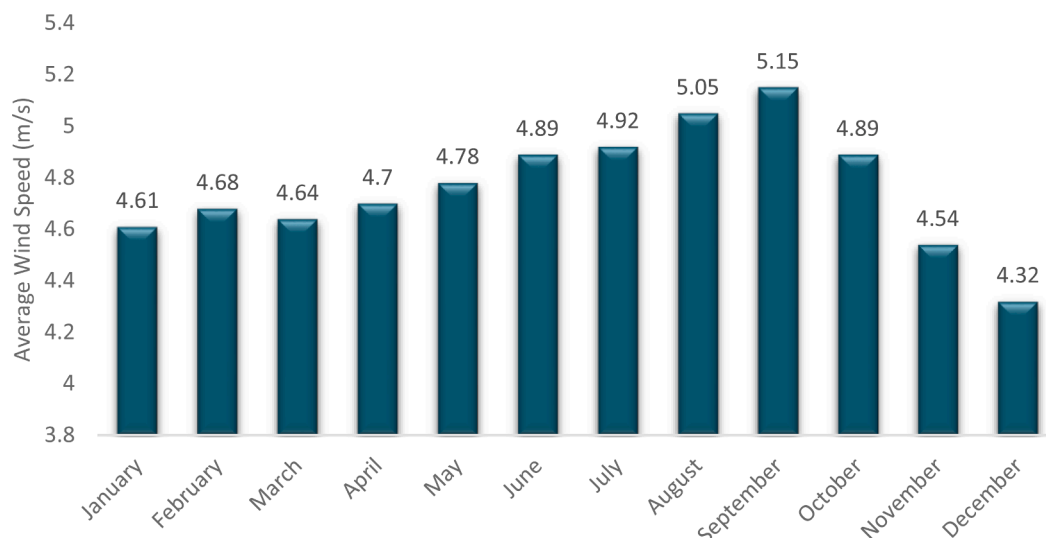


Fig. 7. Average monthly wind speed at the School of Engineering.

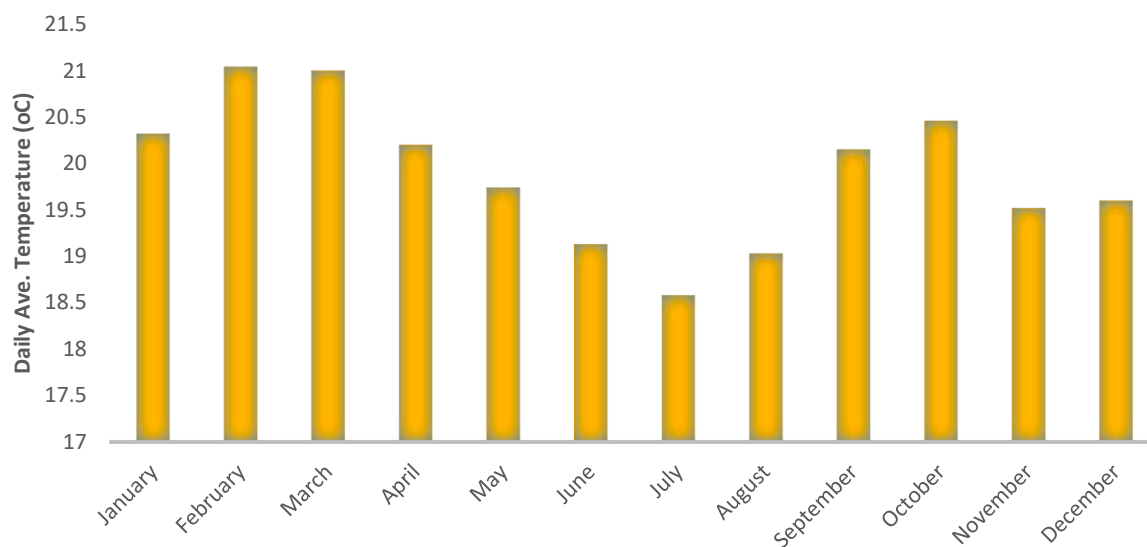


Fig. 8. Monthly temperature variation of School of Engineering.

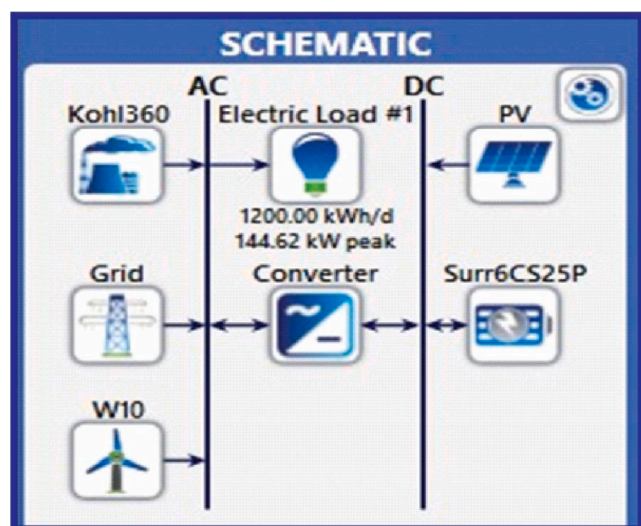


Fig. 9. Schematic of the hybrid system design with added components.

Table 5

Proposed study and optimization scenario.

Proposed System	Scenarios
Grid/PV/Diesel generator system (grid-tied)	A
Grid/PV system (grid-tied)	B
Grid/PV/Wind/Diesel generator system (grid-tied)	C
Grid/PV/Wind System (grid-tied)	D
Grid/PV/Diesel generator/Battery system (grid-tied)	E
Grid/PV/Battery System (grid-tied)	F
Grid/PV/Wind/Diesel generator/Battery system (grid-tied)	G
Grid/PV/Wind/Battery system (grid-tied)	H
PV/Diesel Generator/Battery system (off-grid)	I
PV/Wind/ Diesel generator/Battery system (off-grid)	J

Sensitivity results

The aim of sensitivity analysis is to evaluate the effects of indecisions in the input variables. It answers the question “what if?”. Parameters such as the solar scaled average in kWh/m²/day, wind speed scaled average in m/s, the temperature scaled average in degree Celsius (°C) and the fuel price in USD per liter, can change any time, therefore it is

Table 6

The system optimization components.

PV (kW)	Wind Turbine	Diesel Generator	Battery	Converter	Dispatch Strategy	Grid Maximum Purchase
300 kW	2 × 10 kW	360 kW	300 pieces of 355 W	200 kW	Load Following	999,999 kW

Table 7

Resource input data.

Diesel Fuel price	Solar scaled Average	Temp. scaled average	Wind scaled average
\$1.2/L	5.93 kWh/m ² /day	19.9 °C	4.76 m/s

required to examine the effects these changes may bring to the system specifications and efficiency. This helps to determine the viability and reliability of the project.

The effects of diesel price changes and the solar scaled average on the COE

Fig. 12 shows the effects of changes in the solar scaled average and the price of fuel on the COE for the wind speed of 4.76 m/s and temperature scaled average of 19.9 °C. One can observe that at daily average solar radiation of 4.93 kWh/m²/day and the fuel price of \$0.9/L, the cost of energy is \$0.101/kWh. As the solar radiation increases, the power generated from solar increase and subsequently the cost of

energy. One can notice that as the fuel cost increases, there is no significant change in the COE. This is because HOMER had assumed that the power bought from the grid is enough to supply the energy needed when the PV system generates less energy. The diesel generator is only used if and only if the grid power is off at night. Also, changing the temperature scaled average has no substantial effect on the COE. Wind energy is not included in the selected system and therefore will have no effect.

Similar results were observed for the effects of the diesel price change and the solar scaled average on the NPC, PV power production and the electricity bought from the grid. In general, the effects of changes in diesel price and weather data does not significantly affect the power production and its corresponding cost.

Conclusion and recommendation

The aim of this case study was to assess the technical and economic feasibility of a hybrid renewable energy system to supply adequate, reliable, and affordable power to the school of engineering, University of

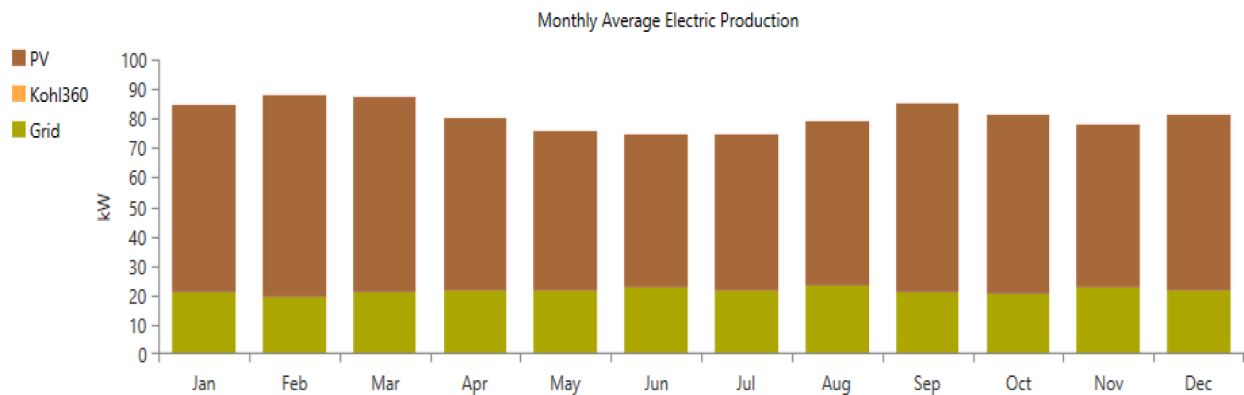


Fig. 10. Monthly average electric power production of the optimal system.

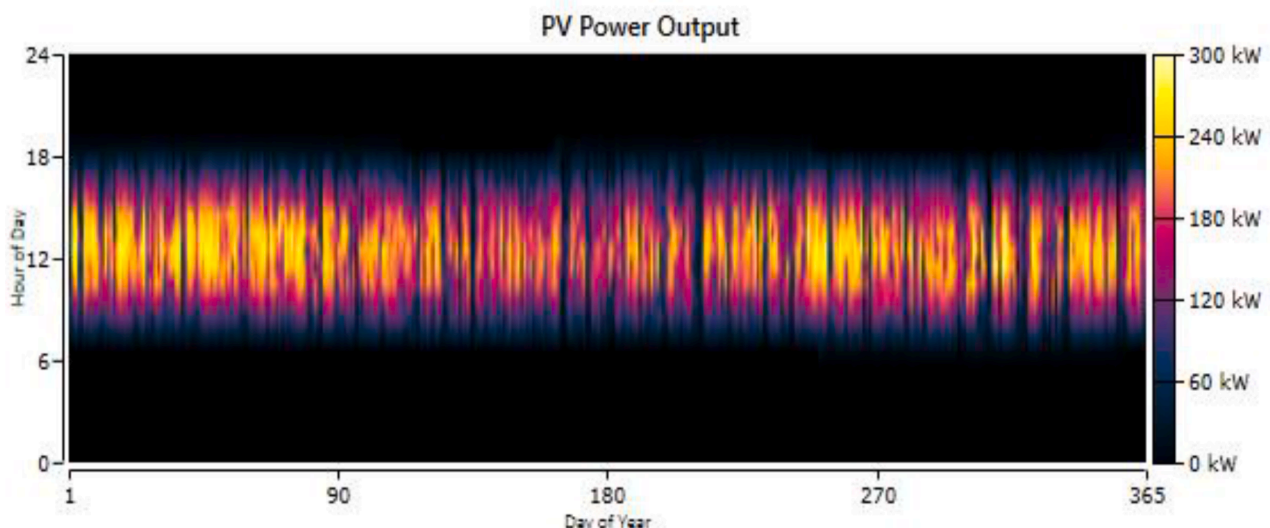
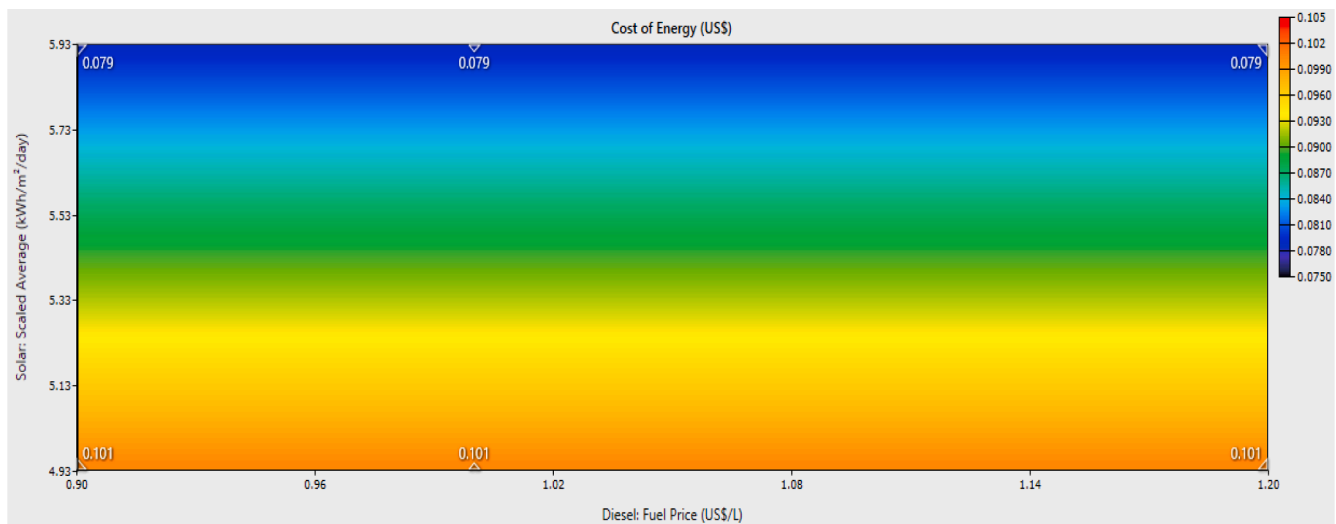


Fig. 11. PV power output of optimal system (kW).

Table 8

Grid power analysis of the optimal system.

Month	Energy Purchased (kWh)	Energy Sold (kWh)	Net Energy Purchased (kWh)	Peak Demand (kW)	Energy Charge	Demand Charge
January	15,616	21,386	-5,770	93.7	US\$556.88	US\$0.00
February	13,053	21,569	-8,516	71.3	US\$22.38	US\$0.00
March	15,588	22,029	-6,441	81.5	US\$474.18	US\$0.00
April	15,528	18,642	-3,114	85.2	US\$868.60	US\$0.00
May	16,149	17,133	-984	83.3	US\$1,174	US\$0.00
June	16,500	14,953	1,546	89.3	US\$1,506	US\$0.00
July	16,082	16,022	60.2	90.2	US\$1,294	US\$0.00
August	17,107	16,886	221	76.8	US\$1,395	US\$0.00
September	15,297	20,132	-4,835	78.6	US\$643.50	US\$0.00
October	15,090	20,238	-5,148	75.1	US\$589.45	US\$0.00
November	16,248	17,219	-970	98.2	US\$1,183	US\$0.00
December	15,953	19,582	-3,629	82.0	US\$840.70	US\$0.00
Annual	188,211	225,791	-37,580	98.2	US\$10,547	US\$0.00

**Fig. 12.** The effects of Diesel Price Changes and the solar scaled average on the COE.

Nairobi. The results have shown that the best technical and economic feasible hybrid renewable energy systems as highlighted is the grid-tied system with component grid, PV and diesel generator. Moreover, this system gives COE of Ksh 7.89, NPC of Ksh 69,512,100.00, Initial capital cost of Ksh 30,264,100.00, a renewable fraction of 71.6%, simple payback period of six years, discounted payback period of 8 years, present worth of Ksh 12,819,300.00, annual worth of Ksh 965,700.00, and an excess electricity generation of 2.48%. The system proves to have reduced the power being purchased from the grid approximately 77% and reduce the electricity bill by approximately 84%. The study also recommends that first, off-grid hybrid system is not cost-effective at the site due to the grid extension which is already in existence. However, grid-tied system will reduce the energy demand from grid to almost zero and therefore is recommended. Second, HOMERTM optimization involves complex simulation, optimization, and sensitivity analyses. Thus, in complex system designs (systems with many components), it is recommended to use high capacity and processor computers. Third, future work should focus on proposing a suitable maneuver and service scheme which can ensure the sustainable operation of the system. Lastly, there should be consideration of the feasibility of biofuels by addressing the likelihood of substituting diesel generator in the system with biofuels which are obtainable from the site.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

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